DO NOW — RECALL

Normalisation to 3NF and indexing

Name: _____ Class: _____ Date: _____

1. What is a transitive dependency?

2. What does an index help with?

3. What does 1NF require?

4. Simplify (A∧B)∨(A∧¬B), stating the Boolean law.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

An attribute depends on key through another non-key attribute.

2. Model answer

Search speed on indexed fields.

3. Model answer

Atomic fields and no repeating groups.

4. Model answer

$A \wedge (B \vee \neg B) = A \wedge 1 = A$.

Vocabulary — know these terms

1NF — Atomic values/no repeating groups.

2NF — No partial dependency on part of composite key.

3NF — No non-key transitive dependency violating key dependency.

index — Search structure over columns.

NEW LEARNING

Read, annotate and explain

Understand

1NF requires atomic values and no repeating groups in a record; 2NF removes dependencies on only part of a composite candidate key.

Apply

3NF eliminates inappropriate transitive dependencies among non-key facts, reducing repeated customer names across orders.

Evaluate

An index accelerates some lookups but uses storage and may slow insert/update operations. Normalisation and indexing solve different problems.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

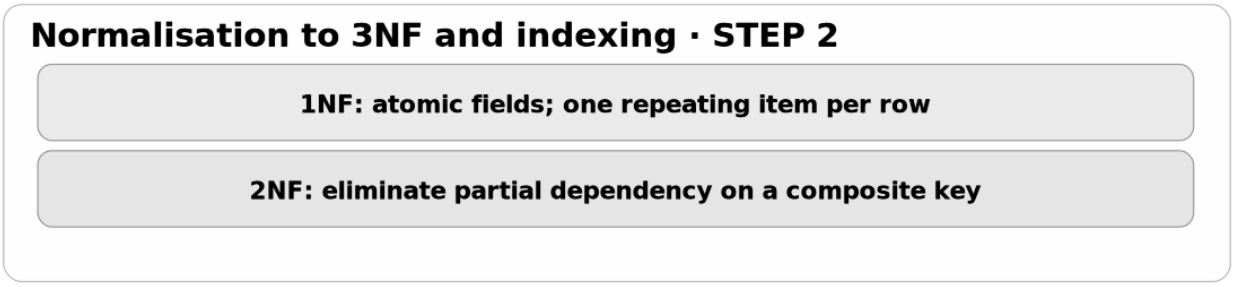
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



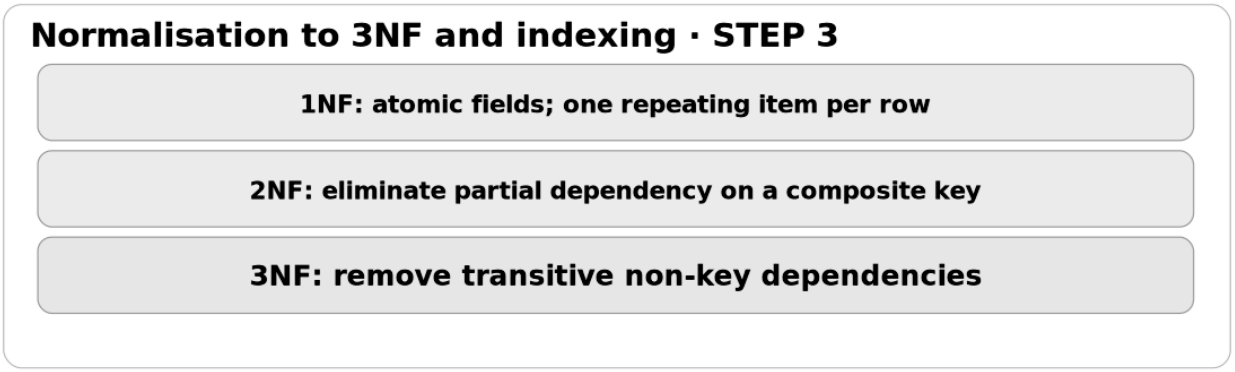
Normalisation to 3NF and indexing · STEP 1

New Learning



Normalisation to 3NF and indexing · STEP 2

New Learning

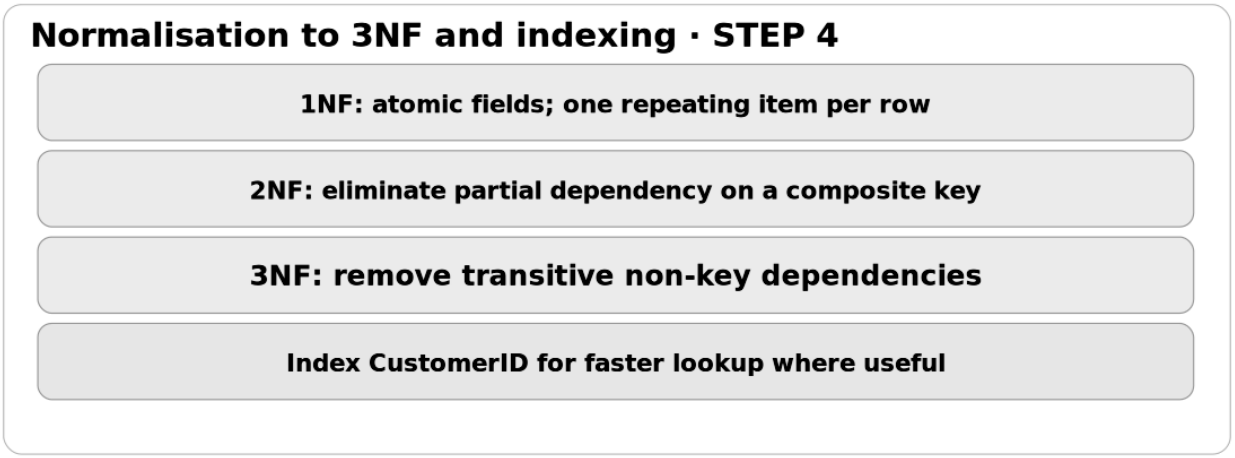


Normalisation to 3NF and indexing · STEP 3

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Normalisation to 3NF and indexing · STEP 4

I DO — WORKED EXAMPLE

Follow the worked method

A table Order(OrderID,CustomerID,CustomerName) repeats a name. Show a 3NF decomposition.

Predict first

Store customer name once in Customer and reference by CustomerID in Order.

Model answer

Explain the reasoning and justify the output.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A table Order(OrderID,CustomerID,CustomerName) repeats a name. Show a 3NF decomposition.

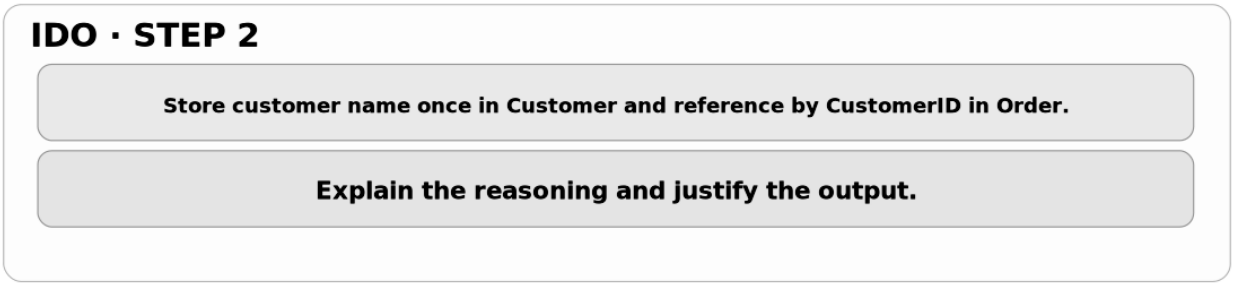
Store customer name once in Customer and reference by CustomerID in Order.



IDO · STEP 1

A table Order(OrderID,CustomerID,CustomerName) repeats a name. Show a 3NF decomposition.

Store customer name once in Customer and reference by CustomerID in Order.



IDO · STEP 2

WE DO — GUIDED PRACTICE

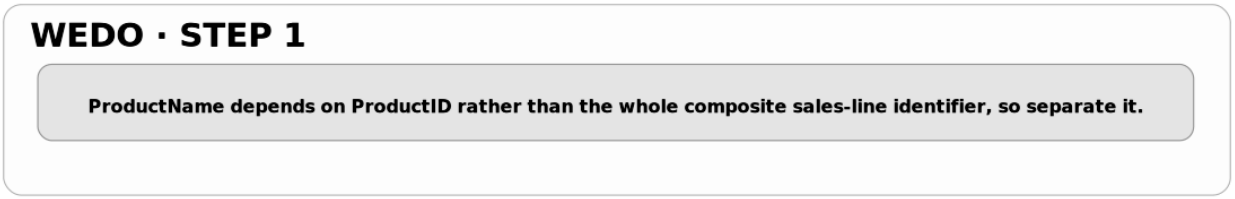
Try → Discuss → Improve

We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

SalesLine(OrderID,ProductID,ProductName,Quantity). Remove product-name dependency.

ProductName depends on ProductID rather than the whole composite sales-line identifier, so separate it.



WEDO · STEP 1

WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

Guided 1: Predict first ProductName depends on ProductID rather than the whole composite sales-line identifier, so separate it.

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

IMPROVE — Write one precise correction or stronger explanation in your own answer.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

State 1NF,2NF,3NF in one sentence each. [6]

Question 2

Order(OrderID, CustomerID, CustomerName). Explain the redundancy and split into two tables. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Explain why an index on StudentName can speed SELECT but also incur costs. [6]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

Normalising to 3NF means the database never needs indexes.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

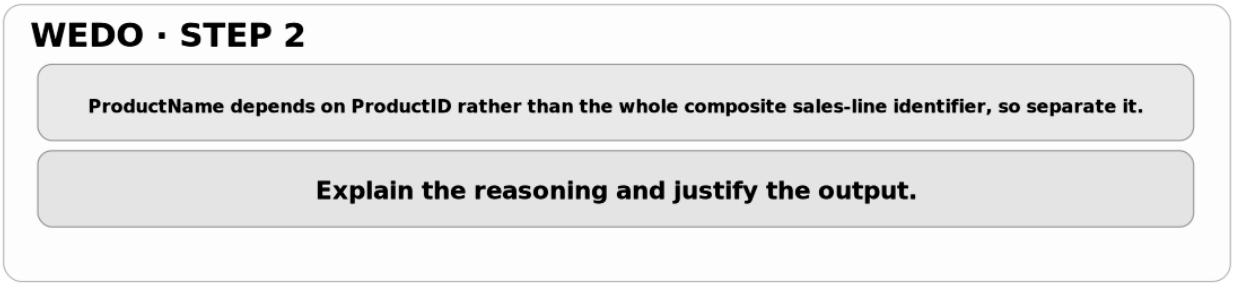
Next step: revisit the model or a relevant revision-guide section if needed.

We Do Check · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

SalesLine(OrderID,ProductID,ProductName,Quantity). Remove product-name dependency.

ProductName depends on ProductID rather than the whole composite sales-line identifier, so separate it.



WEDO · STEP 2

WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

Guided 1 — original CHECK

Model answer

Explain the reasoning and justify the output.

IMPROVE: Explain one correction to your own reasoning.